

Shweta Misra

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Summary

Machine Learning Engineer with 4+ years of experience **designing, fine-tuning, and deploying scalable AI/ML systems**, with proven impact at Tendmill in optimizing procurement using **GenAI**, building **RAG-enabled GPT agents**, and developing scalable ML-powered search and parsing systems. Skilled in **end-to-end ML engineering**-from model development and backend integration to **scalable data pipelines**-leveraging tools like **PyTorch, PostgreSQL, and Elastic** to deliver production-ready solutions. Adept at driving cross-functional alignment in fast-paced environments, with a strong track record of combining **technical innovation and business impact** across startups and enterprise consulting.

Work Experience

Consultant (Data and AI)

Netlight Consulting

May 2025 - Aug 2025

- **Data Integration into ML Knowledge Retrieval System:** Integrated Zendesk data into a RAG system powering Netlight's internal ChatGPT interface by building a scalable ELT pipeline using Snowflake and dbt
- **Scaling optimization for World Food Program's Logistics Planning Tool:** Collaborated with engineering teams to diagnose and reproduce scaling issues in logistics optimization models hosted on Palantir Foundry
- **Designed & Developed Scalable Data Transformation Pipelines in Palantir:** Developed and deployed parallelized data transformation pipelines in Palantir Foundry for the WFP's scenario planning tool, ensuring efficient execution under tight compute constraints and integrating robust test harnesses for reliability

Machine Learning Engineer

Tendmill AB (acquired by Matilda FoodTech - Jan'25)

Sep 2022 - Apr 2025

- **Optimizing Procurement with GenAI:** Designed and fine-tuned Transformer-based GenAI models (BERT, GPT) on food procurement data, achieving 98% precision, in requirements to article matching, enhancing overall procurement efficiency by 20%
- **Article Classification:** Developed RAG-enabled GPT agents to categorize over 200,000 articles into a hierarchical taxonomy and built supporting backend services to manage and utilize categorized articles
- **AI-Enabled Search Optimization:** Implemented vector search and prompt-engineered pipelines using Qdrant, Elastic, PSQL and LLM APIs, doubling response speed for AI-driven food article recommendation engine
- **Semi-structure Text Parsing:** Spearheaded the development of LLM-driven system to extract structured requirements from semi-structured procurement documents, replacing rigid regex-based parser with fine-tuned BERT model to improve parsing accuracy by 50%
- **Cross-Functional Collaboration in fast-paced startup environment:** Partnered with business and engineering teams to align AI-driven features with strategic objectives, ensuring delivery of time-critical business impact

Business Technology Associate Consultant

ZS Associates

Jul 2018 - Jul 2021

- **Data Analytics for Forecasting:** Developed statistical forecasting models for oncology treatment, reducing forecasting cycle time by 200% and optimizing business decision-making.
- **Automation & Data Pipelines:** Led Designing and maintenance of automated Tableau reporting solutions, reducing manual effort and operational reporting costs by 70%
- **Performance Enablement & Strategy:** Worked in tandem with client and internal teams to create intuitive dashboards to communicate impact of regional resource distribution, enabling more granular strategy-making across therapy areas

Skills

Programming & Analytics Tools

Python, JavaScript, TypeScript, Node.js, React, Flask, Tableau, Git, Cursor

AI & ML

GPT, BERT, RAG, Transformers, PyTorch, OpenAI & Anthropic APIs, Scikit-learn

Data Engineering & Cloud

PostgreSQL, Qdrant, Docker, Palantir Foundry, AWS (ECS, S3), ElasticSearch

Education

Master of Science in Computer Science (Data Science Track)

KTH Royal Institute of Technology, Sweden

Aug 2021 - Aug 2023

Bachelor of Engineering in Computer Science

University of Pune, India

Aug 2014 - Jun 2018

Projects & Publications

Nested Noun Phrase Detection using Transformers (LREC-COLING 2024, Torino, Italy) 🏆

KTH

Feb 2023 - Aug 2023

Designed recursive Transformer-based models achieving 95% accuracy in parsing complex multilingual syntax

Article Clustering for Data-Driven Product Strategy

Tendmill AB (acquired by Matilda FoodTech - Jan'25)

Jun 2022 - Aug 2022

Developed clustering models to identify challenges in food procurement categories, improving operational efficiency

Semi-supervised learning on LSTMs for Automatic Speech Recognition (ASR)

KTH

Feb 2022 - May 2022

Student-teacher training of LSTM models using semi supervised learning to annotate phonemes in audio samples from the TIMIT corpus